

Supplementary Information

From Clinical Signal Recovery to Evidence-Qualified Interpretation of Prostate Cancer
Pathology Foundation Models

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Scope and reading rule

This Supplementary Information preserves the complete evidence needed to reproduce or challenge the main manuscript’s target-specific alignment states. It does not enlarge the claim ceiling. In particular, recoverability means that a target can be predicted from a frozen representation by a patient-disjoint or locked external probe; it does not mean that a disease-prediction head functionally uses the target. Grade alone has a separate internal exploratory locked-head erasure analysis; it is not an external, indispensable, mechanistic, or clinical-increment result. No clinician trust, reliance, diagnostic-performance, or clinical-utility outcome was evaluated.

Abbreviations. AR, androgen receptor; AUROC, area under the receiver operating characteristic curve; BCR, biochemical recurrence; C-index, concordance index; CONCH, CONTRastive learning from Captions for Histopathology; DFS, disease-free survival; H&E, hematoxylin and eosin; IBS, integrated Brier score; ISUP, International Society of Urological Pathology; PANDA, Prostate cANcer graDe Assessment; PFI, progression-free interval; PFS, progression-free survival; PTEN, phosphatase and tensin homolog; SPOP, speckle type BTB/POZ protein; TCGA-PRAD, The Cancer Genome Atlas prostate adenocarcinoma cohort.

S1. Cohort membership, target definition, and analysis unit

The main study combined five resources for complementary purposes rather than pooling them into one cohort. NADT-Prostate supplied source grade and phenotype probes. PANDA supplied Karolinska and Radboud case images for locked transfer. PRECISE supplied an additional session-level grade evaluation. TCGA-PRAD supplied molecular, site, and endpoint audits. LEOPARD supplied the source recurrence analysis. Patient, case-image, session, and slide units were never treated as interchangeable.

Supplementary Table S1 preserves the detailed clinical, reference, and metric orientation removed from the concise Main Introduction. It makes explicit that every reported estimate used a paired target reference while external and functional reference availability differed across axes.

Supplementary Table S1: Clinical-target, paired-reference, and metric primer. The metric follows the target type rather than a common accuracy scale. A positive representation result means that a probe can recover or rank the recorded target; it does not mean that H&E replaces the clinical or molecular reference assay.

Axis	What is measured clinically or biologically	Paired reference and availability	How the study reads the target
Grade/ISUP	Ordinal architectural differentiation; grade is not tumor amount.	Pathologist or source grade in NADT, PANDA, PRECISE, and TCGA; compatible external references were available.	Ordinal rank target; Spearman ρ , with 0 as the null.
Tumor phenotype/content	Presence or fraction of tumor-like tissue; not a molecular subtype or aggressiveness score.	NADT tumor fraction and PANDA grade-derived tumor/benign status; independent same-patient truth was unavailable in the TCGA BCR frame.	Continuous or binary target; Spearman ρ or AUROC, with AUROC 0.5 as chance.
PTEN loss	Binary loss of a tumor-suppressor gene signal measured by molecular copy number.	TCGA-PRAD copy-number reference; compatible external transport data were unavailable.	Binary molecular target; AUROC. A positive result is a histology–molecular association.
SPOP mutation	Binary genomic mutation state, not directly visible as a named H&E structure.	TCGA-PRAD genomic reference was available; lack of a qualified signal was not caused by a missing target label.	Binary molecular target; AUROC. Near-chance performance does not prove absence of biological importance.
AR activity	Continuous androgen-receptor-regulated transcriptional activity; not an immunostain or binary mutation.	TCGA-PRAD transcriptional activity score; compatible external transport data were unavailable.	Continuous molecular target; Spearman ρ , with grade-adjusted change in R^2 for conditional information.
Recurrence	Time from treatment to a defined return or progression event.	LEOPARD and TCGA time-to-event references; event definitions were available but non-equivalent.	Censored time-to-event target; C-index for ordering and IBS for prediction error.

Supplementary Table S2 records why the five resources and the correlated setting audit could not be pooled into one denominator or one validation claim.

Supplementary Table S2: Study resources, analysis frames, and roles in signal qualification. Counts identify the unit used by each component and must not be summed because patients, slides, folds, and settings can overlap.

Resource or component	Target and saved analysis frame	Role in the alignment design
NADT-Prostate	Grade and phenotype; 39 patients	Source-cohort probe fitting and patient-level evaluation
PANDA	Grade: 565 Karolinska and 572 Radboud case images; phenotype: 469 and 483 tumor cases	Cross-cohort transfer without target-cohort refitting
PRECISE	Grade; 17 evaluable imaging sessions	Additional resource with a session-level analysis unit
TCGA-PRAD	Molecular analyses: 273 patients; recurrence transfer: 270 patients; paired recurrence increment: 153 complete cases; FM6 frame: 392 patients and 80 events	Molecular, clinical-increment, site, endpoint, and scoped locked-head audits
LEOPARD	Post-hoc recurrence risk score fitted in the source cohort	Transfer without refitting to the TCGA-PRAD target frames
Correlated setting audit	Six targets; 72 configurations, 360 seed cells, and 390 paired contrasts	Encoder–scale–tile–seed sensitivity, not independent validation

Supplementary Table S3 is the reading key for all subsequent analyses: it defines each target’s biological layer, analysis unit, metric, and the strongest permitted interpretation before any numerical result is shown.

Supplementary Table S3: Registered target definitions and interpretation boundaries. The unit and metric specify how each target was evaluated; the permitted-interpretation column is the claim ceiling and does not imply disease prediction or functional use by a downstream head.

ID	Layer	Target	Unit	Metric	Permitted interpretation
T01	Morphologic	Gleason or ISUP grade	Patient, case image, or session	Spearman rho	Grade information is recoverable and directionally transports
T02	Morphologic	Tumor phenotype or content	Patient or case image	Spearman rho or AUROC	Morphology-linked phenotype is recoverable and transports
T03	Molecular	PTEN loss	Patient	AUROC	Within-cohort PTEN-related information is recoverable
T04	Molecular	SPOP mutation	Patient	AUROC	Unsupported in the frozen primary design
T05	Molecular	AR activity	Patient	Spearman rho	Positive pooled within-cohort alignment
T06	Outcome	Post-hoc recurrence risk	Patient	C-index and paired increment	Exploratory endpoint-conditioned alignment

Supplementary Table S4 expands the Main evidence-state summary by showing why the six prostate-cancer axes are biologically non-interchangeable and by separating representation evidence from contribution to the BCR judgment.

Supplementary Table S4: Six prostate-cancer axes and the evidence stage reached in this study. “Functional contribution” refers specifically to a locked BCR-head intervention, not to whether the target is clinically or biologically relevant.

Target	Meaning in prostate cancer	Signal in the frozen representation	Functional contribution to the AI judgment
Grade/ISUP	Architectural differentiation and aggressiveness	Best-qualified recoverability and transport; CONCH and Virchow internal ISUP directions	Internal exploratory BCR-head sensitivity demonstrated
Tumor phenotype/content	Presence or amount of tumor-like tissue, not aggressiveness	Strong recoverability and external morphology transport	Currently blocked: same-cohort independent tumor-content truth is unavailable
PTEN loss	Loss of a tumor-suppressor gene	Recoverable but conditional beyond grade and without external transport	Feasible follow-up, not run; requires a new overlap, event-power, fold, and erasure protocol
AR activity	Activity of the androgen-receptor pathway	Positive but conditional beyond grade and across sites	Feasible follow-up, not run; requires a new overlap, event-power, fold, and erasure protocol
SPOP mutation	Molecular mutation state	No qualified signal in the frozen primary design	Not qualified for head-use testing because recoverability was unsupported
Recurrence	Patient outcome influenced by tumor biology, treatment, and follow-up	Endpoint- and reference-sensitive image-risk association	Not applicable as an input-feature erasure target; requires external prognostic validation

S2. Frozen encoder, scale, sampling, aggregation, and fold contracts

CONCH and Virchow weights remained frozen. Tissue-qualified tiles were embedded and aggregated to slides; patient-unit endpoints then used patient aggregation. The correlated sensitivity program crossed two encoders, two physical-scale choices per encoder, three tile budgets, and five sampling seeds. Within-cohort probes used patient-disjoint folds. The external morphology probes were applied without target-cohort refitting.

Continuous marker-family targets used standardized ridge regression with the saved fixed alpha grid. Binary TCGA-PRAD screening used standardized logistic regression with fixed regularization and

balanced class weighting. The frozen NADT phenotype probe used for PANDA transfer retained its source specification: `LogisticRegression(C=1.0)`, 2,000 maximum iterations, and no class weighting. Within-cohort evaluation used five grouped folds, with all slides from one patient confined to one fold. Each out-of-fold prediction therefore came from a model that had not seen that patient. Native refits and held-out site analyses remained separate evaluation types and did not substitute for locked transfer.

S3. Target-specific estimands and nulls

Ordinal and continuous targets used Spearman correlation with null zero. Binary targets used AUROC with null 0.5. Time-to-event risk used C-index with null 0.5. Conditional increment nulls were zero change from the registered clinical reference. Metric values were not pooled across target types.

Ridge regression is linear regression with a penalty on large coefficients, used here to stabilize high-dimensional correlated embeddings. Logistic regression maps features to a binary class probability. Spearman correlation measures rank preservation without requiring a linear relationship; AUROC is the probability that a randomly chosen positive patient is ranked above a negative patient; and the survival C-index is the corresponding ordering measure for comparable patient pairs under censoring.

The saved 17-test discovery family used two-sided Spearman tests for continuous targets and two-sided Mann–Whitney tests for binary targets, with Benjamini–Hochberg false-discovery-rate control at $\alpha = 0.05$. The Mann–Whitney test compares score distributions without assuming normality. Grade-stratified permutation tests shuffled patient labels and refitted the full nested procedure to estimate the no-increment reference distribution. Qualification repeats in another cohort or encoder were not added to the discovery family.

S4. Complete primary estimates and uncertainty accounting

The NADT grade and phenotype rows retained patient-bootstrap intervals but not complete valid/undefined replicate accounting. PANDA and PRECISE aggregate rows did not retain intervals. TCGA molecular primary rows retained interval endpoints; binary event counts were not retained in those saved primary rows. These unavailable quantities were not reconstructed.

Supplementary Table S5 reports the complete primary recoverability and external morphology-transport estimates. Because its rows use different estimands, a Spearman correlation, AUROC, and their denominators must be read within their own row and must not be compared as if they shared a common scale.

Supplementary Table S5: Primary known-target alignment estimates. Metrics, analysis units, and denominators are row-specific; intervals are 95 percent patient-bootstrap intervals where retained, and unavailable intervals were not reconstructed.

Target	Cohort	Metric	Unit	n	Estimate	Interval
Gleason	NADT	Spearman rho	Patient	39	0.478	0.170 to 0.712
Phenotype	NADT	Spearman rho	Patient	39	0.800	0.625 to 0.909
Gleason	PANDA	Spearman rho	case image	565	0.354	unavailable
Gleason	PANDA	Spearman rho	case image	572	0.519	unavailable
Phenotype	PANDA	AUROC	case image	565	0.786	unavailable
Phenotype	PANDA	AUROC	case image	572	0.871	unavailable
Gleason	PRECISE	Spearman rho	Session	17	0.865	unavailable
PTEN	TCGA-PRAD	AUROC	Patient	273	0.632	0.560 to 0.706
SPOP	TCGA-PRAD	AUROC	Patient	273	0.519	0.408 to 0.627
AR	TCGA-PRAD	Spearman rho	Patient	273	0.195	0.074 to 0.310

S5. External transport details

The NADT grade direction persisted in the two PANDA institutions and the PRECISE resource. The NADT phenotype probe retained discrimination in both PANDA institutions. These are locked-transfer or separate-resource results, but their units and missing uncertainty remain row-specific. No compatible external cohort was available for PTEN, SPOP, or AR activity.

S5A. Internal ISUP functional-sensitivity bridge and detector boundary

The two encoder analyses used identical outer patient-disjoint folds. Within each outer training fold, 64 principal components were fitted, standardized ridge regression learned the ISUP direction, and a ridge-penalized Cox model learned BCR risk. Regularization was selected from 0.1, 1, 10, 100, and 1,000 using training data only. The normalized ISUP-probe coefficient defined the targeted direction. The primary test held the fitted Cox coefficients fixed while projecting that direction out of held-out representations. One hundred random directions removed the same amount of representation variance, 20 label-permuted directions provided a negative control, and the one-sided empirical p value compared the targeted change with the matched-random changes. A secondary analysis refitted the Cox head after erasure. Patient-bootstrap intervals used 2,000 resamples and ISUP-label permutation used 200 permutations.

The FM6 internal frame contained 392 TCGA-PRAD patients, 80 BCR events, and 27,968 paired whole-tissue tiles. The frozen CONCH and Virchow patient representations used the same five outer patient-disjoint folds. ISUP ridge probes achieved out-of-fold Spearman correlations of 0.615 and 0.658, while the corresponding full BCR heads achieved C-indices of 0.627 (0.553–0.694) and 0.632 (0.564–0.691).

For the fixed-head test, targeted removal of the ISUP-correlated direction reduced C-index by 0.041 (0.011–0.073) for CONCH and 0.023 (0.007–0.040) for Virchow. Each targeted change exceeded 100 variance-matched random-direction controls with one-sided empirical $p = 0.0099$. After the BCR head was refitted on the erased representation, the corresponding differences were 0.014 (–0.006–0.035) and 0.010 (–0.003–0.022), so both intervals included zero. The limited ISUP-plus-AI versus ISUP-only comparison was –0.000 (–0.039–0.039) for CONCH and –0.032 (–0.088–0.025) for Virchow. This comparison does not establish absence of clinical increment; it only failed to establish increment in the scoped internal frame.

The manuscript-closing clean rerun used the unchanged 392-patient/80-event membership, paired embeddings, five folds, seed, endpoints, preprocessing, regularization grids, resampling counts, controls, and output schema. Before execution, SHA-256 hashes for 20 protocol-defined nonvolatile analysis outputs were retained as the baseline. All 20 regenerated hashes matched exactly, and the ISUP recoverability, full-head C-index, fixed-head erasure, matched-random comparison, refit-after-erasure, limited ISUP-only comparison, and R/A/U/T decisions were unchanged. Timestamp and device metadata were treated as volatile and were not used to claim byte identity. This audit demonstrates exact computational reproducibility of the stored internal experiment, not an independent sample, institution, or endpoint replication.

The functional analysis was not tumor-restricted. The remediated detector’s opened SICAP secondary test had AUROC 0.919, sensitivity 0.875, and specificity 0.810. At the unchanged threshold, newly held-out PANDA sensitivity was 0.562 in Karolinska and 0.679 in Radboud, despite specificity of 0.943 and 0.913. SICAP specificity is accepted as internal technical evidence, but independent-domain sensitivity remains unresolved. Consequently, tumor-specific functional use, external functional transport, and strong mechanistic alignment remain unsupported.

S5B. Locked site-heldout and LEOPARD functional transport

The site-heldout analysis used the seven TCGA tissue-source sites with at least 20 subjects and five BCR events, totaling 289 unique evaluation subjects and 69 events. Every site was excluded from its training partition. Cross-site patient pairs were excluded from the primary stratified C-index. CONCH produced a full-head C-index of 0.612 (0.524–0.696) and targeted $\Delta_{use} = 0.048$ (−0.005–0.102); its target interval included zero. Virchow produced 0.588 (0.514–0.665) and $\Delta_{use} = 0.033$ (0.007–0.060), respectively. Both targeted effects exceeded matched-random controls after Holm correction, but only Virchow passed all prespecified encoder criteria. The family therefore closed as partial encoder-specific site-heldout evidence rather than replicated internal transport.

The external reanalysis applied one TCGA-only scaler, 64-component PCA transform, Cox head, and ISUP-correlated direction per encoder to 508 LEOPARD patients with 87 BCR events. All 32,512 outcome-blind sampled crops were identical across encoders by coordinate and decoded crop hash. CONCH yielded a full-head C-index of 0.517 (0.449–0.590) and $\Delta_{use} = 0.038$ (0.014–0.066; Holm $p = 0.0198$). Virchow yielded 0.457 (0.386–0.528) and $\Delta_{use} = 0.002$ (−0.008–0.011; Holm $p = 0.2277$). CONCH failed the required full-head discrimination criterion, while Virchow failed both head validity and targeted sensitivity. Neither encoder passed external whole-tissue functional transport. Two exact reruns reproduced all nonvolatile aggregate output hashes.

LEOPARD patients and institution were independent of TCGA, but LEOPARD lacked ISUP and treatment covariates and its outcomes had been accessed in an earlier directionally reversed analysis in this research program. The test is therefore a newly locked independent-patient external reanalysis, not a prospectively untouched confirmation. It cannot establish external ISUP recoverability, endpoint equivalence, tumor-specific mechanism, clinical increment, indispensability, or strong H2. The prespecified negative or inconclusive gate was retained without retuning.

Supplementary Table S6: Locked site-heldout and independent-patient functional-transport tests. C-index intervals and targeted erasure deltas use 2,000 patient-bootstrap draws. Passing required valid full-head discrimination, a positive targeted-delta interval, matched-random significance after Holm correction, and the family-specific sample gate.

Frame	Encoder	Full-head C-index (95% CI)	Target delta (95% CI)	Gate
TCGA site-heldout	conch	0.612 (0.524 to 0.696)	+0.048 (−0.005 to +0.102)	Fail or inconclusive
LEOPARD external	conch	0.517 (0.449 to 0.590)	+0.038 (+0.014 to +0.066)	Fail or inconclusive
TCGA site-heldout	virchow	0.588 (0.514 to 0.665)	+0.033 (+0.007 to +0.060)	Pass
LEOPARD external	virchow	0.457 (0.386 to 0.528)	+0.002 (−0.008 to +0.011)	Fail or inconclusive

S6. Conditional increment and confounder audits

PTEN and AR increments were computed in nested patient-disjoint analyses. Neither encoder’s stored interval excluded zero for PTEN AUROC increment or AR R^2 increment. These results limit grade-independent interpretation but do not prove that grade exhausts the molecular signal.

Supplementary Table S7 makes the conditional comparison explicit: each delta is the held-out image-augmented model minus its grade-only reference, so a positive value favors adding the image-derived score.

Supplementary Table S7: Held-out molecular-target increments beyond grade. Delta is the image-augmented model minus the grade reference on untouched outer-fold patients; positive values favor adding the image score. Intervals are 95 percent patient-bootstrap intervals.

Target	Encoder	Metric	n	Delta	Interval
PTEN	CONCH	AUROC	273	+0.019	-0.025 to +0.068
AR	CONCH	R2	273	+0.004	-0.036 to +0.042
PTEN	Virchow	AUROC	273	+0.035	-0.013 to +0.087
AR	Virchow	R2	273	+0.019	-0.015 to +0.051

S7. Site characteristics and unresolved transport

The pooled AR patient-level estimate was positive. The six site-specific estimates used a slide-level effect with patient-cluster bootstrap, and every saved interval crossed zero. Site was entangled with tissue source, scanner, stain, preparation, and case mix; absent metadata were not inferred. For SPOP, one site was single-class and therefore had undefined AUROC, and the other stored site intervals included or touched chance.

Supplementary Table S8 retains every AR and SPOP site row, including the undefined single-class SPOP row. The site abbreviations are source identifiers rather than verified scanner or staining protocols, and the differing analysis units are displayed explicitly.

Supplementary Table S8: Complete AR and SPOP site audits. AR effects are slide-level with patient-cluster intervals; SPOP uses patient AUROC. The CH SPOP row is undefined because no positive event was present. Bootstrap accounting reports undefined draws out of 2,000 where applicable.

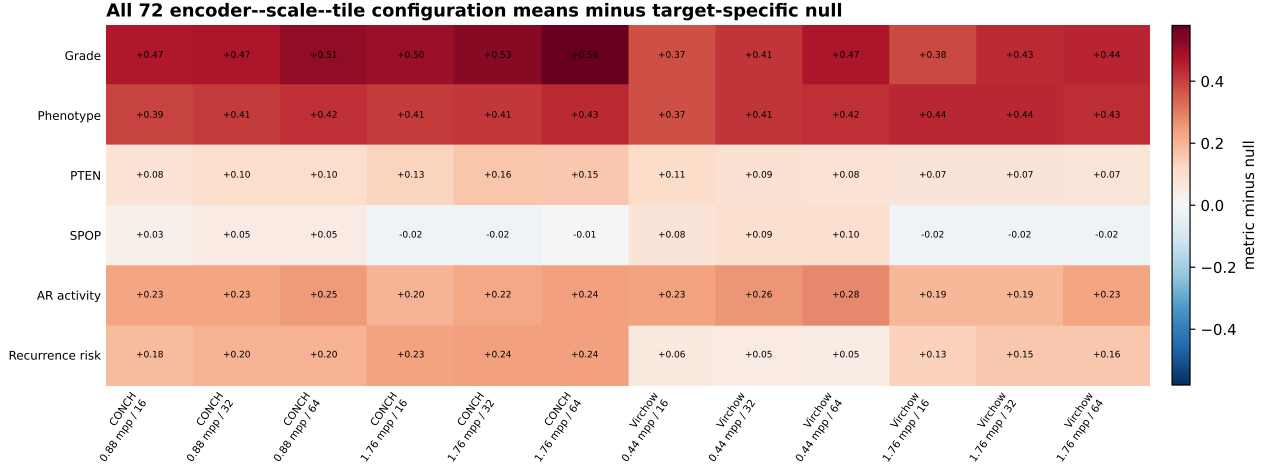
Target	Site	Metric	Patients	Events	Estimate	Interval	Accounting
AR	CH	Spearman rho	23	not applicable	-0.128	-0.561 to 0.350	Event count not applicable
AR	EJ	Spearman rho	59	not applicable	0.127	-0.121 to 0.363	Event count not applicable
AR	G9	Spearman rho	27	not applicable	0.195	-0.258 to 0.598	Event count not applicable
AR	HC	Spearman rho	47	not applicable	0.217	-0.072 to 0.471	Event count not applicable
AR	KK	Spearman rho	31	not applicable	0.265	-0.087 to 0.558	Event count not applicable
AR	YL	Spearman rho	22	not applicable	0.227	-0.135 to 0.540	Event count not applicable
AR	Pooled	Spearman rho	273	not applicable	0.195	0.074 to 0.310	Event count not applicable
SPOP	CH	AUROC	23	0	undefined	unavailable	Undefined: single class
SPOP	EJ	AUROC	59	7	0.613	0.418 to 0.798	0/2,000
SPOP	G9	AUROC	27	2	0.700	0.423 to 0.923	234/2,000
SPOP	HC	AUROC	47	3	0.758	0.500 to 0.939	92/2,000
SPOP	KK	AUROC	31	6	0.573	0.273 to 0.840	2/2,000
SPOP	YL	AUROC	22	4	0.625	0.282 to 1.000	31/2,000

S8. Full encoder–scale–tile–seed sensitivity program

The complete grid comprised 72 configurations and 360 correlated seed cells. The paired contrast table comprised 390 seed-matched comparisons. These settings reused cohorts, patients, and folds. Their intervals describe tile-sampling variation and their contrasts describe analytic sensitivity; neither is population uncertainty or independent replication.

For each configuration, five seed-level values were summarized by mean, sample standard deviation, range, chance-or-worse count, and a Student-*t* interval. Paired comparisons kept the seed fixed when subtracting scale, encoder, or tile-budget settings. These intervals quantify repeated tile-sampling variation only; they are not patient-bootstrap confidence intervals and were not used to revise labels, folds, or evidence thresholds.

Supplementary Figure S1 shows every configuration mean relative to its target-specific null. Supplementary Table S9 then condenses those same correlated settings into ranges and null-straddling counts. The Main manuscript displays the target-level range summary; Supplementary Figure S2 retains the more detailed seed-matched setting contrasts. These views describe the same sensitivity program and are not independent replications.



Supplementary Figure S1: Complete encoder–scale–tile configuration grid. Each cell is the mean of five sampling seeds minus the target-specific metric null. Columns retain encoder, physical scale, and tile budget; rows retain target identity. Values reuse cohorts and folds and therefore describe correlated setting sensitivity rather than independent validation.

Supplementary Table S9: Observed ranges across 12 correlated encoder–scale–tile configurations. Mean and range summarize the five-seed configuration values; null-straddling is the number of configurations whose observed seed range crossed the target-specific null, out of 12. These are sensitivity summaries, not independent validations.

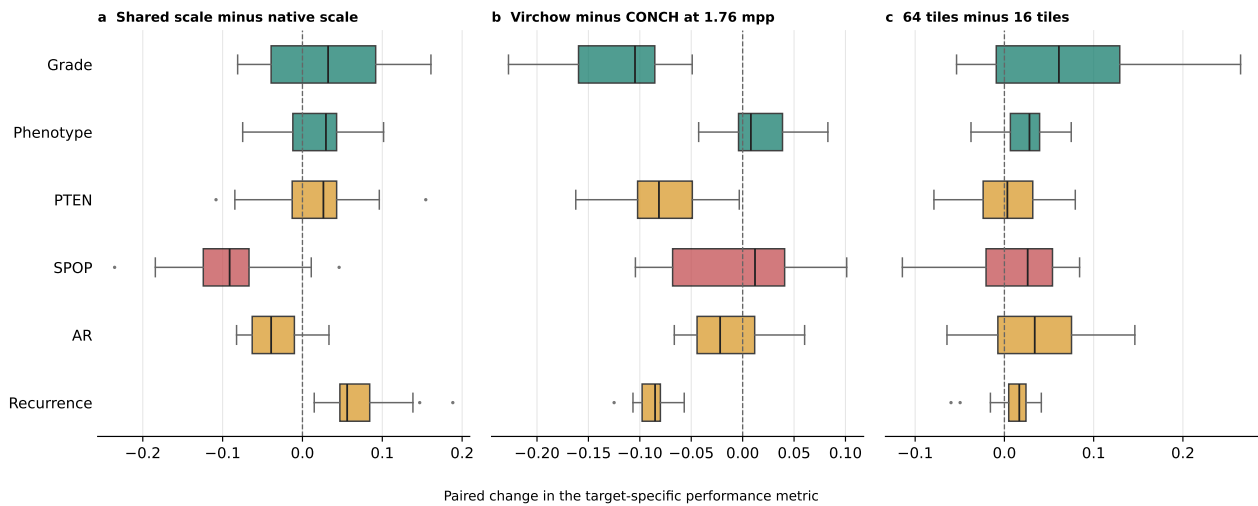
Target	Metric	Null	Mean	Range	Null-straddling
Grade	Patient Spearman rho	0.0	0.464	0.271 to 0.646	0/12
Phenotype	Patient AUROC	0.5	0.915	0.845 to 0.965	0/12
PTEN	Patient AUROC	0.5	0.602	0.519 to 0.717	0/12
SPOP	Patient AUROC	0.5	0.523	0.348 to 0.679	8/12
AR activity	Patient Spearman rho	0.0	0.230	0.136 to 0.297	0/12
Recurrence risk	Patient C-index	0.5	0.657	0.481 to 0.755	1/12

The full contrast-level source contains 390 seed-matched comparisons. To expose their direction and spread without treating them as independent observations, Supplementary Table S10 summarizes every target within each registered contrast family. Median and range describe technical sensitivity; null-crossing counts describe a change in the target-specific chance relation, not a population-level hypothesis test.

Supplementary Table S10: Complete summary of the 390 seed-matched setting contrasts. Delta direction is setting B minus setting A in the named contrast.

Contrast	Target	Pairs	Median delta	Range	Null crossing	Exact ties
Shared 1.76 mpp - native scale	Grade	30	+0.032	-0.081 to +0.161	0	0
Shared 1.76 mpp - native scale	Phenotype	30	+0.029	-0.075 to +0.102	0	0
Shared 1.76 mpp - native scale	PTEN	30	+0.026	-0.108 to +0.154	0	0
Shared 1.76 mpp - native scale	SPOP	30	-0.091	-0.235 to +0.046	19	0
Shared 1.76 mpp - native scale	AR activity	30	-0.039	-0.083 to +0.033	0	0
Shared 1.76 mpp - native scale	Recurrence risk	30	+0.056	+0.015 to +0.188	1	0
Virchow - CONCH at 1.76 mpp	Grade	15	-0.105	-0.228 to -0.049	0	0
Virchow - CONCH at 1.76 mpp	Phenotype	15	+0.008	-0.043 to +0.083	0	1
Virchow - CONCH at 1.76 mpp	PTEN	15	-0.081	-0.162 to -0.003	0	0
Virchow - CONCH at 1.76 mpp	SPOP	15	+0.012	-0.104 to +0.101	4	0
Virchow - CONCH at 1.76 mpp	AR activity	15	-0.022	-0.066 to +0.060	0	0
Virchow - CONCH at 1.76 mpp	Recurrence risk	15	-0.085	-0.125 to -0.057	0	0
64 tiles - 16 tiles	Grade	20	+0.061	-0.054 to +0.265	0	0
64 tiles - 16 tiles	Phenotype	20	+0.028	-0.037 to +0.075	0	1
64 tiles - 16 tiles	PTEN	20	+0.003	-0.079 to +0.079	0	0
64 tiles - 16 tiles	SPOP	20	+0.026	-0.114 to +0.084	5	0
64 tiles - 16 tiles	AR activity	20	+0.034	-0.064 to +0.146	0	0
64 tiles - 16 tiles	Recurrence risk	20	+0.017	-0.060 to +0.041	0	0

Supplementary Figure S2 visualizes the same paired contrasts and preserves their direction: shared scale minus native scale, Virchow minus CONCH at the shared scale, and 64 minus 16 tiles. A positive value therefore has meaning only within the named comparison.



Supplementary Figure S2: Seed-matched encoder, physical-scale, and tile-budget contrasts. Each box summarizes paired performance changes for one target while preserving the sampling seed. **a**, Shared physical scale minus encoder-native scale. **b**, Virchow minus CONCH at the shared 1.76-micrometre-per-pixel scale. **c**, 64 minus 16 tiles per slide. All comparisons reuse cohorts and folds and do not define a universally preferable setting.

S9. Endpoint hierarchy and paired outcome increments

Reconstructed with-tumor recurrence, strict recurrence-only, TCGA-CDR PFS, TCGA-CDR DFS, and official PFI retained separate identifiers. On the 270-patient target frame, reconstructed recurrence had 57 events and official PFI had 42. Paired increment analyses used 153 complete cases, with 30 and 15 events respectively. All official-PFI paired C-index and IBS intervals included zero.

Supplementary Table S11 defines the endpoint IDs, their roles, metrics, and the non-equivalence boundary that prevents a result under one recurrence definition from being silently relabeled as another.

The transferred image-risk score was fitted with a Cox model in LEOPARD and applied to TCGA-PRAD without target-cohort refitting. Its principal-component dimension sweep was exploratory and was not nested inside an outer evaluation. Paired TCGA comparisons used the same complete-case patients and bootstrap draw for the reference and augmented models. C-index improvement was augmented minus reference; integrated Brier score improvement was reference minus augmented, so positive values favored augmentation under both definitions. The Cox model estimates relative event hazard under censoring, while integrated Brier score summarizes time-dependent prediction error and is lower when predictions are better.

The comparison names in Supplementary Tables S12 and S13 have the following meaning. “Grade + image vs grade” and “full clinical + image vs full clinical” test augmentation by the transferred image risk score; here the full clinical reference contains grade, age, PSA, pathological T stage, and surgical-margin status. “Image vs grade” and “image vs full clinical” compare non-nested reference choices rather than incremental models. In the separate prespecified hierarchy, M4 contains grade, PSA, pathological T stage, surgical-margin status, and tissue-source site, whereas M5 adds the image risk score. Thus “full clinical + site + image vs full clinical + site” is the M5–M4 comparison. These labels describe model contents, not evidence that the image score adds clinical value.

Supplementary Table S11: Registered endpoint hierarchy and non-equivalence boundaries. Primary rows define known-target recoverability, secondary rows test conditional or horizon-specific questions, and exploratory rows audit recurrence definitions. Each row keeps its own metric and boundary; endpoint IDs are not interchangeable outcomes.

ID	Hierarchy	Target or endpoint	Metric	Role	Boundary
E01	Primary	Continuous or ordinal marker target	Spearman rho	Known-target recoverability	Not a disease decision
E02	Primary	Binary marker target	AUROC	Known-target recoverability	Not a molecular diagnosis
E03	Secondary	PTEN or AR beyond grade	Delta AUROC or delta R2	Conditional uniqueness beyond grade	Not evidence of functional use
E04	Exploratory	Reconstructed recurrence with tumor	C-index, time-dependent AUROC, and IBS	Recurrence alignment	Not official PFI
E05	Exploratory	Strict recurrence only	C-index and time-dependent AUROC	Sensitivity analysis	Not official PFI
E06	Exploratory	TCGA-CDR PFS	C-index, time-dependent AUROC, and landmark AUROC	Sensitivity analysis	Not relabeled as PFI
E07	Exploratory	TCGA-CDR DFS	C-index, time-dependent AUROC, and landmark AUROC	Sensitivity analysis	Not relabeled as PFI
E08	Exploratory	Official TCGA-CDR PFI	C-index and endpoint concordance	Outcome alignment	Not equivalent to reconstructed recurrence
E09	Secondary	Three- and five-year landmarks	Landmark C-index or AUROC	Sensitivity analysis	Not equivalent to a time-to-event analysis

Supplementary Tables S12 and S13 retain the full frozen-risk and paired comparison frames. The

paired rows use identical patients and bootstrap draws within an endpoint. The two endpoint tables must not be pooled because their event definitions and event counts differ.

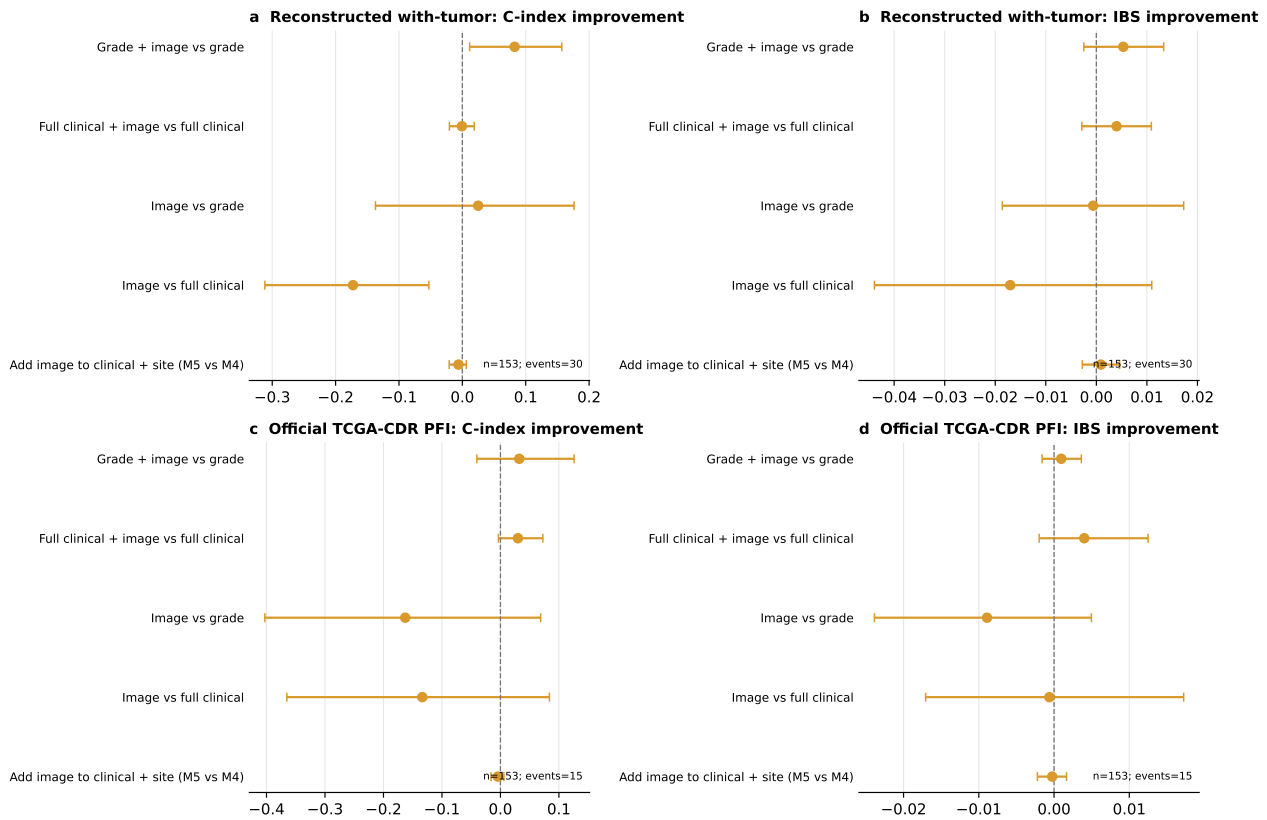
Supplementary Table S12: Complete frozen-risk and paired comparisons for the reconstructed recurrence-with-tumor endpoint. Paired rows use the same patients and bootstrap draws. C-index delta is augmented or first model minus reference; IBS delta is reference minus augmented or first model, so positive values favor the augmented or first model. Intervals are 95 percent patient-bootstrap intervals; accounting reports undefined draws out of 2,000.

Comparison	Metric	n	Events	Estimate	Interval	Undefined / 2,000
Frozen image-risk score	C-index	270	57	+0.673	+0.587 to +0.759	0/2,000
Full clinical + image vs full clinical	C-index	153	30	-0.001	-0.020 to +0.019	0/2,000
Full clinical + image vs full clinical	IBS (0.5–5 years)	153	30	+0.004	-0.003 to +0.011	0/2,000
Grade + image vs grade	C-index	153	30	+0.083	+0.012 to +0.157	0/2,000
Grade + image vs grade	IBS (0.5–5 years)	153	30	+0.005	-0.002 to +0.013	0/2,000
Image vs full clinical	C-index	153	30	-0.173	-0.312 to -0.053	0/2,000
Image vs full clinical	IBS (0.5–5 years)	153	30	-0.017	-0.044 to +0.011	0/2,000
Image vs grade	C-index	153	30	+0.025	-0.137 to +0.176	0/2,000
Image vs grade	IBS (0.5–5 years)	153	30	-0.001	-0.019 to +0.017	0/2,000
Add image to clinical + site (M5 vs M4)	C-index	153	30	-0.006	-0.021 to +0.006	0/2,000
Add image to clinical + site (M5 vs M4)	IBS (0.5–5 years)	153	30	+0.001	-0.003 to +0.005	0/2,000

Supplementary Table S13: Complete frozen-risk and paired comparisons for official TCGA-CDR PFI. Paired rows use the same patients and bootstrap draws. C-index delta is augmented or first model minus reference; IBS delta is reference minus augmented or first model, so positive values favor the augmented or first model. Intervals are 95 percent patient-bootstrap intervals; accounting reports undefined draws out of 2,000.

Comparison	Metric	n	Events	Estimate	Interval	Undefined / 2,000
Frozen image-risk score	C-index	270	42	+0.586	+0.482 to +0.689	0/2,000
Full clinical + image vs full clinical	C-index	153	15	+0.030	-0.003 to +0.073	0/2,000
Full clinical + image vs full clinical	IBS (0.5–5 years)	153	15	+0.004	-0.002 to +0.013	0/2,000
Grade + image vs grade	C-index	153	15	+0.032	-0.040 to +0.126	0/2,000
Grade + image vs grade	IBS (0.5–5 years)	153	15	+0.001	-0.002 to +0.004	0/2,000
Image vs full clinical	C-index	153	15	-0.134	-0.365 to +0.084	0/2,000
Image vs full clinical	IBS (0.5–5 years)	153	15	-0.001	-0.017 to +0.017	0/2,000
Image vs grade	C-index	153	15	-0.163	-0.403 to +0.069	0/2,000
Image vs grade	IBS (0.5–5 years)	153	15	-0.009	-0.024 to +0.005	0/2,000
Add image to clinical + site (M5 vs M4)	C-index	153	15	-0.004	-0.016 to +0.006	0/2,000
Add image to clinical + site (M5 vs M4)	IBS (0.5–5 years)	153	15	-0.000	-0.002 to +0.002	0/2,000

Supplementary Figure S3 plots the paired intervals from the two tables on a common positive-favors-first-or-augmented convention; endpoint-specific panels remain separate because their event definitions and counts differ.



Supplementary Figure S3: Complete paired outcome-comparison family across non-equivalent endpoints. Rows compare the image score with grade or fuller clinical references, or compare nested clinical models, on the same patients and bootstrap draws. **a,b**, Reconstructed with-tumor C-index and integrated-Brier-score improvements. **c,d**, Corresponding official TCGA-CDR PFI comparisons. Positive values favor the first, augmented, or more complete model under the registered delta definition.

S10. Multiplicity, undefined draws, and power boundaries

The saved discovery family contained 17 tests with Benjamini–Hochberg false-discovery-rate control. External and representation checks of an existing target were qualification analyses. Missing intervals, missing event counts, single-class sites, and underpowered endpoint rows remained explicit. No non-significant result was interpreted as biological absence.

Where saved sources supported it, uncertainty used 95% percentile patient-bootstrap intervals. The bootstrap resampled patients with replacement and recomputed the statistic; AR site analyses instead resampled patients as clusters around a slide-level effect. Paired survival contrasts used identical draws for both models. Requested, valid, and undefined replicate counts were reported when retained and otherwise marked unavailable. No missing or unevaluable outcome was converted to a negative label, and multiple imputation was not introduced retrospectively.

The promoted derivative evidence retains the family-level multiplicity contract but does not contain a complete row-level table of all raw and adjusted p values. Those values were not reconstructed from prose. Supplementary Table S14 instead reports every retained availability and bootstrap-accounting state across the promoted figure-source tables, making clear which limitations arise from missing intervals, event counts, single-class sites, or legacy replicate accounting.

Supplementary Table S14: Row-level availability and uncertainty-accounting states retained in the promoted evidence. Counts are source-table rows, not patients or independent tests; unavailable quantities and undefined single-class estimates were preserved rather than reconstructed.

Evidence component	Availability/accounting state	Rows
External transport	Interval not saved	5
External transport	Replicate accounting not saved	2
Molecular alignment	Bootstrap accounted	5
Molecular alignment	Event count not applicable	2
Molecular alignment	Event count not recorded in source	4
Molecular alignment	Primary metric undefined	1
Conditional and site audits	Event count not applicable	9
Conditional and site audits	Event count not recorded in source	2
Outcome	Bootstrap accounted	22

S11. Claim–evidence and numeric lineage

Each main claim is connected to a PBV source claim, a target identifier, required evidence axes, and a functional-use status. Each reported headline number is linked to a source ID, semantic key, field, and display rule. The full machine-readable records are `provenance/claim_evidence_matrix.csv` and `provenance/numeric_qa_mapping.csv`.

Supplementary Table S15 exposes the manuscript claim ID, source-claim lineage, registered target IDs, evidence state, and functional-use status together. It is the final check that a representation-level result has not been promoted to a locked-head or clinical claim.

Supplementary Table S15: Claim–evidence and functional-use boundary. A identifiers denote this manuscript’s claims, C identifiers denote promoted source claims, and T identifiers denote registered targets. The state summarizes the qualified evidence, whereas functional use records whether a locked-head intervention was tested; recoverability alone does not establish head use.

Claim	Source claims	Targets	State	Functional use
A01	C01; C08	T01; T02; T03; T04; T05; T06	Descriptive framework	Not tested
A02	C02	T01; T02	Transportable	Not tested
A03	C03	T03	Context-sensitive	Not tested
A04	C05	T05	Context-sensitive	Not tested
A05	C04	T04	Unsupported in frozen design	Not tested
A06	C06; C07	T06	Context-sensitive	Not tested
A07	C08	T01; T02; T03; T04; T05; T06	Descriptive framework	Not tested
A08	QFM-FM6-SUMMARY; QFM-FM6-CONTRASTS	T01	Internal exploratory	Internal exploratory only
A09	QFM-FM6-CONTRASTS	T01	Context-sensitive	Internal exploratory only
A10	none	T01; T02; T03; T04; T05; T06	Deferred follow-up	Not claimed
A11	QFM-FM6-SITE-HELDOUT; QFM-FM6-LEOPARD-EXTERNAL	T01	external fail or inconclusive	external tested not qualified

S12. Reproducibility and known limitations

The manuscript builder first verifies every source path against its registered byte count and SHA-256 hash. It then generates new QFM-owned figures and tables from those sources. No PBV-generated PDF, PNG, or TeX table is copied. The PBV project remains owner of the frozen analyses, while QFM owns the derivative alignment narrative.

For the FM6 extension specifically, clean-rerun handoff required 20/20 nonvolatile output hashes, unchanged headline numbers, and unchanged gates. Both encoders retained R=whole-tissue re-coverability, A=internal disease association, and U=exploratory whole-tissue functional sensitivity. Site-heldout transport was encoder-specific; the locked LEOPARD family failed or remained inconclusive for both encoders, so T did not pass and strong H2 remained prohibited. These gate labels describe evidence scope rather than a sequence of clinical validation levels.

Unequal availability of external target-matched cohorts limits comparison between target classes. The correlated grid is not independent validation; site effects are not causal acquisition effects; and recurrence endpoints are not interchangeable. The grade analysis includes a locked BCR head and target-erasure experiment, but only as whole-tissue internal exploratory evidence. Its detector did not pass the independent-domain sensitivity gate, refitted heads could compensate, and the limited ISUP-only comparison was not a sufficient clinical-increment study. The present paper therefore establishes candidate shared interpretive coordinates and scoped internal grade sensitivity, not complete explanation of AI decisions. Residual or unknown-feature marker discovery is reserved for a separate study.